QUALITY INFORMATION DOCUMENT

For OMI_EXTREME_WMF_MEDSEA

Issue: 1.2

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2022/06/16	All	New OMI QUID	Emanuela Clementi	Anna Chiara Goglio (PQ resp)
1.1	2023/06/16	All	Extension of time series to year 2021	Emanuela Clementi	Anna Chiara Goglio (PQ resp)
1.2	2025/05/28	All	Extension of the time series, Setup of operational temporal extension and renaming of the product name	Lucia Gualtieri	Emanuela Clementi

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1) INTRODUCTION

a) Product Name

OMI_EXTREME_WMF_MEDSEA_area_averaged_mean

b) Indicator Evaluation Method

The Mediterranean water mass formation rates are evaluated in 4 areas as defined in the Ocean State Report Issue 2 (OSR2, von Schuckmann et al., 2018) Section 3.4 (Simoncelli and Pinardi, 2018) and shown in Figure 1: (1) the Gulf of Lions for the Western Mediterranean Deep Waters (WMDW); (2) the Southern Adriatic Sea Pit for the Eastern Mediterranean Deep Waters (EMDW); (3) the Cretan Sea for Cretan Intermediate Waters (CIW) and Cretan Deep Waters (CDW); (4) the Rhodes Gyre, the area of formation of the so-called Levantine Intermediate Waters (LIW) and Levantine Deep Waters (LDW).
Annual water mass formation rates have been computed using daily mixed layer depth estimates (density criteria $\Delta\sigma$ = 0.01 kg/m3, 10 m reference level) considering the annual maximum volume of water above mixed layer depth with potential density within or higher the specific thresholds specified in Table 1 then divided by seconds per year.
Annual mean values are provided using the Mediterranean 1/24° eddy resolving reanalysis (Escudier et al. 2020, 2021).

Region	Water Mass	Lon [°E]	Lat [°N]	Density Threshold [kg/m3]
1. Gulf of Lion	WMDW	3.5–6.5	41–42.6	>29.10
2. South Adriatic Pit	EMDW	17–19	41–42.5	>29.10
3. Cretan Sea	CDW/CIW	23–26.5	35.5–37	CDW >29.2 CIW [29.1 - 29.2]
4. Rhode Gyre	LDW/LIW	26.3–30	34–36.1	LDW >29.2. LIW [28.95 -29.10]

Table 1. Details of the four areas of deep water formation (see boxes in Figure 2) considered for water mass formation rate computation and the potential density thresholds used.

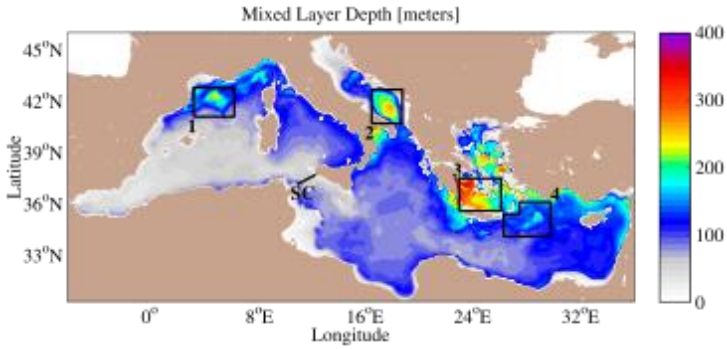

Figure 1: From OSR.2 Figure 3.4.1 (Simoncelli and Pinardi, 2018). Mixed Layer Depth [m] and black boxes representing areas where deep water formation is known to occur: (1) the Gulf of Lions for the Western Mediterranean Deep Waters (WMDW); (2) the Southern Adriatic Sea Pit for the Eastern Mediterranean Deep Waters (EMDW); (3) the Cretan Sea for the Cretan Intermediate Waters (CIW) and the Cretan Deep Waters (CDW); (4) the Rhodes Gyre for the Levantine Intermediate Waters (LIW) and the Levantine Deep Waters (LDW).

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
MEDSEA_MULTIYEAR_PHY_006_004	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-MED-QUID-006-004.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-MED-PUM-006-004.pdf

2) SCIENTIFIC VALIDATION

a) Mediterranean Water Mass Formation Rates

More details and full scientific evaluation can be found in the Copernicus Marine Ocean State Report issue 2 (OSR2, von Schuckmann et al., 2018) Section 3.4 (Simoncelli and Pinardi, 2018). With respect to the results shown in OSR2, an updated higher resolution Mediterranean Sea Reanalysis has been used to evaluate the present OMI.
The scientific validation of this index relies on the quality of the Mediterranean Sea Reanalysis and is enforced showing correspondence between the occurrence of the largest formation rates with respect to events reported in literature.
The formation of intermediate and deep water masses is one of the most important processes occurring in the Mediterranean Sea, being a component of its general overturning circulation. This circulation varies at interannual and multidecadal time scales and it is composed of an upper zonal cell (Zonal Overturning Circulation) and two main meridional cells in the western and eastern Mediterranean (Pinardi and Masetti 2000).
This OMI aims to monitor the main water mass formation events using the eddy resolving Mediterranean Sea Reanalysis (Escudier et al. 2020, 2021) and considering Pinardi et al. (2015) and Simoncelli and Pinardi (2018) as references for the methodology. The Mediterranean Sea Reanalysis is able to reproduce both Eastern Mediterranean Transient and Western Mediterranean Transition phenomena and catches the principal water mass formation events reported in the literature. This will permit constant monitoring of the open ocean deep convection process in the Mediterranean Sea and a better understanding of the multiple drivers of the general overturning circulation at interannual and multidecadal time scales.
Figure 2 shows the time variability of deep and intermediate water formation events characterized by a deep mixed layer depth occurring in four areas of the basin: (1) the Gulf of Lions for the Western Mediterranean Deep Waters (WMDW); (2) the Southern Adriatic Sea Pit for the Eastern Mediterranean Deep Waters (EMDW); (3) the Cretan Sea for the Cretan Intermediate Waters (CIW) and the Cretan Deep Waters (CDW); (4) the Rhodes Gyre for the Levantine Intermediate Waters (LIW) and the Levantine Deep Waters (LDW). Time spans from 1987 to the year preceding the current one [-1Y], operationally extended yearly.

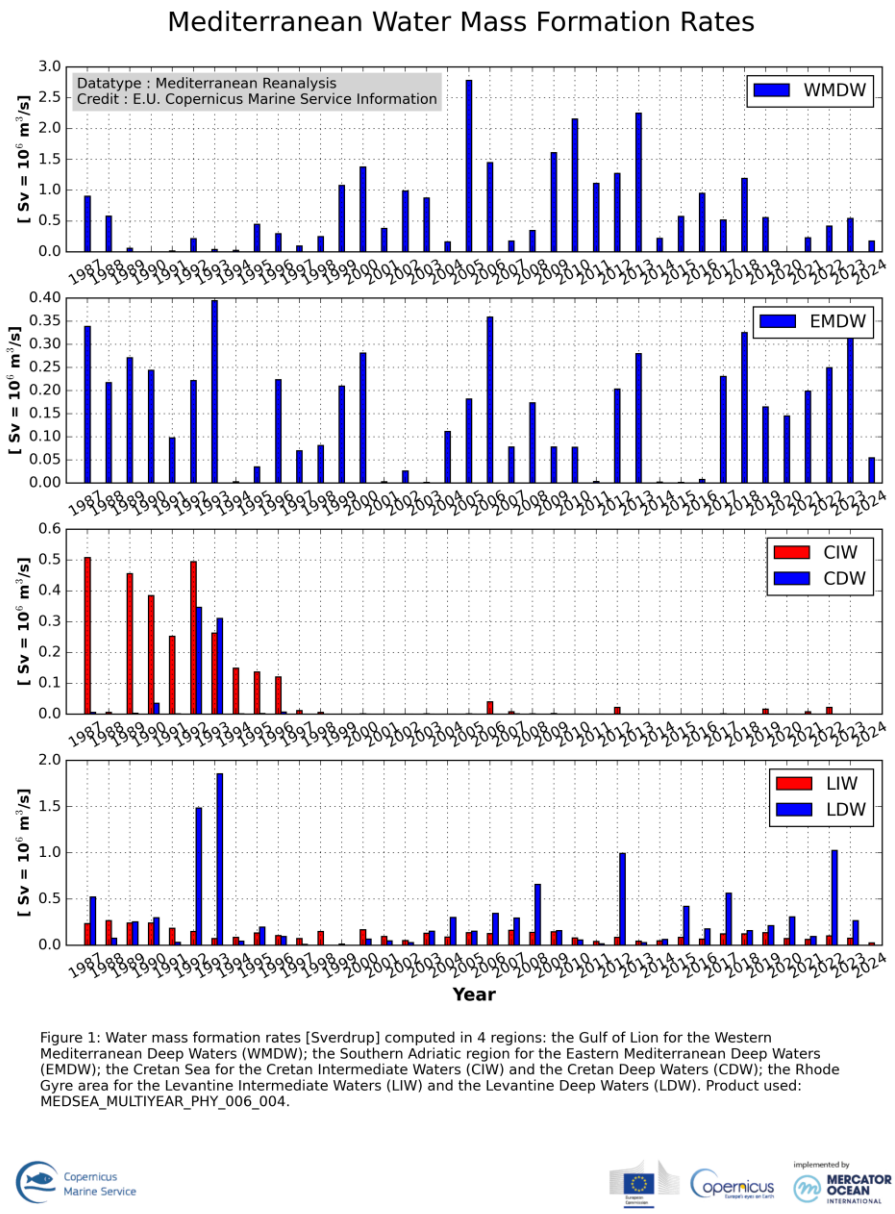

Figure 1: Water mass formation rates [Sverdrup] computed in 4 regions: the Gulf of Lion for the Western Mediterranean Deep Waters (WMDW); the Southern Adriatic region for the Eastern Mediterranean Deep Waters (EMDW); the Cretan Sea for the Cretan Intermediate Waters (CIW) and the Cretan Deep Waters (CDW); the Rhode Gyre area for the Levantine Intermediate Waters (LIW) and the Levantine Deep Waters (LDW). Product used: MEDSEA_MULTIYEAR_PHY_006_004.

Figure 1: Water mass formation rates [Sverdrup] computed in 4 regions (black boxes in Figure 1): in the Gulf of Lion for the Western Mediterranean Deep Waters (WMDW); in the Southern Adriatic region for the Eastern Mediterranean Deep Waters (EMDW); in the Cretan Sea for the Cretan Intermediate Waters (CIW) and the Cretan Deep Waters (CDW); in the Rhode Gyre area for the Levantine Intermediate Waters (LIW) and the Levantine Deep Waters (LDW). Product used: MEDSEA_MULTIYEAR_PHY_006_004.

The Western Mediterranean Deep Water (WMDW) formation events in the Gulf of Lion appears to be larger after 1999 consistently with Schroeder et al. (2006, 2008) related to the Eastern Mediterranean Transient event. This modification of WMDW after 2005 has been called Western Mediterranean Transition. WMDW formation events are consistent with Somot et al. (2016) and the event in 2009 is also reported in Houpert et al. (2016).

The Eastern Mediterranean Deep Water (EMDW) formation in the Southern Adriatic Pit region displays a period of water mass formation between 1988 and 1993, in agreement with Pinardi et al. (2015), in 1996, 1999 and 2000 as documented by Manca et al. (2002). Weak deep water formation in winter 2006 is confirmed by observations in Vilibić and Šantić (2008). An intense deep water formation event is detected in 2012-2013 (Gačić et al., 2014). Last years are characterized by large events starting from 2017 (Mihanovic et al., 2021).
Cretan Intermediate Water formation rates present larger peaks between 1989 and 1993 with the ones in 1992 and 1993 composing the Eastern Mediterranean Transient phenomena. The Cretan Deep Water formed in 1992 and 1993 is characterised by the highest densities of the entire period in accordance with Velaoras et al. (2014).
The Levantine Deep Water formation rate in the Rhode Gyre region presents the largest values between 1992 and 1993 in agreement with Kontoyiannis et al. (1999).

3) REFERENCES

Escudier R., Clementi E., Cipollone A., Pistoia J., Drudi M., Grandi A., Lyubartsev V., Lecci R., Aydogdu A., Delrosso D., Omar M., Masina S., Coppini G., Pinardi N. 2021. A High Resolution Reanalysis for the Mediterranean Sea. Frontiers in Earth Science, Vol.9, pp.1060, DOI:10.3389/feart.2021.702285.

Escudier, R., Clementi, E., Omar, M., Cipollone, A., Pistoia, J., Aydogdu, A., Drudi, M., Grandi, A., Lyubartsev, V., Lecci, R., Cretí, S., Masina, S., Coppini, G., & Pinardi, N. (2020). Mediterranean Sea Physical Reanalysis (CMEMS MED-Currents) (Version 1) set. Copernicus Monitoring Environment Marine Service (CMEMS). https://doi.org/10.25423/CMCC/MEDSEA_MULTIYEAR_PHY_006_004_E3R1

Gačić, M., Civitarese, G., Kovačević, V., Ursella, L., Bensi, M., Menna, M., et al. 2014. Extreme winter 2012 in the Adriatic: an example of climatic effect on the BiOS rhythm. Ocean Sci. 10, 513–522. doi: 10.5194/os-10-513-2014

Houpert, L., de Madron, X.D., Testor, P., Bosse, A., D'Ortenzio, F., Bouin, M.N., Dausse, D., Le Goff, H., Kunesch, S., Labaste, M., et al. 2016. Observations of open-ocean deep convection in the northwestern Mediterranean Sea: seasonal and inter- annual variability of mixing and deep water masses for the 2007-2013 period. J Geophys Res Oceans. 121:8139–8171. doi:10.1002/ 2016JC011857.

Kontoyiannis, H., Theocharis, A., Nittis, K. 1999. Structures and characteristics of newly formed water masses in the NW levantine during 1986, 1992, 1995. In: Malanotte-Rizzoli P., Eremeev V.N., editor. The eastern Mediterranean as a laboratory basin for the assessment of contrasting ecosys- tems. NATO science series (series 2: environmental secur- ity), Vol. 51. Springer: Dordrecht.

Manca, B., Kovacevic, V., Gačić, M., Viezzoli, D. 2002. Dense water formation in the Southern Adriatic Sea and spreading into the Ionian Sea in the period 1997–1999. J Mar Sys. 33/ 34:33–154.

Mihanović, H., Vilibić, I., Šepić, J., Matić, F., Ljubešić, Z., Mauri, E., Gerin, R., Notarstefano, G., Poulain, P.-M.. 2021. Observation, preconditioning and recurrence of exceptionally high salinities in the Adriatic Sea. Frontiers in Marine Science, Vol. 8, https://www.frontiersin.org/article/10.3389/fmars.2021.672210

Schroeder, K., Gasparini, G.P., Tangherlini, M., Astraldi, M. 2006. Deep and intermediate water in the western Mediterranean under the influence of the eastern Mediterranean transient. Geophys Res Lett. 33. doi:10. 1028/2006GL02712.

Schroeder, K., Ribotti, A., Borghini, M., Sorgente, R., Perilli, A., Gasparini, G.P. 2008. An extensive western Mediterranean deep water renewal between 2004 and 2006. Geophys Res Lett. 35(18):L18605. doi:10.1029/2008GL035146.

Simoncelli, S. and Pinardi, N. 2018. Water mass formation processes in the Mediterranean sea over the past 30 years. In: Copernicus Marine Service Ocean State Report, Issue 2, Journal of Operational Oceanography, 11:sup1, s13–s16, DOI: 10.1080/1755876X.2018.1489208.

Somot, S., Houpert, L., Sevault, F., Testor, P., Bosse, A., Taupier-Letage, I., Bouin, M.N., Waldman, R., Cassou, C., Sanchez-Gomez, E., et al. 2016. Characterizing, modelling and under- standing the climate variability of the deep water formation in the North-Western Mediterranean Sea. Clim Dyn. 1–32. doi:10.1007/s00382-016-3295-0.

Velaoras, D., Krokos, G., Nittis, K., Theocharis, A. 2014. Dense intermediate water outflow from the Cretan Sea: a salinity driven, recurrent phenomenon, connected to thermohaline circulation changes. J Geophys Res Oceans. 119:4797–4820. doi:10.1002/2014JC009937.

Vilibić, I., Šantić, D. 2008. Deep water ventilation traced by Synechococcus cyanobacteria. Ocean Dyn 58:119–125. doi:10.1007/s10236-008-0135-8.

Von Schuckmann K. et al. (2018) Copernicus Marine Service Ocean State Report, Journal of Operational Oceanography, 11:sup1, S1-S142, DOI: 10.1080/1755876X.2018.1489208